
Research Interests

I am broadly interested in Natural Language Processing, Machine Learning, and Information Retrieval. My recent research has focused on code generation using large language models (LLMs). Specifically, I aim to build and evaluate CodeLLMs to assist software engineering.

Education

- 2022 – 2026 **PhD, Natural Language Processing**, The University of Edinburgh, Scotland, UK.
Research Unit: Institute for Language, Cognition and Computation (ILCC)
Research Areas: Program Synthesis, Code Generation, Neuro-Symbolic Reasoning, Compositional Generalisation, Large Language Models
Advisor: Prof. [Shay Cohen](#) and Prof. [Antonio Vergari](#)
- 2020 – 2021 **M.Sc. by Research, Informatics**, The University of Edinburgh, Scotland, UK.
Research Unit: Institute for Language, Cognition and Computation (ILCC)
Research Areas: Unsupervised Parsing, Knowledge Graph Completion
Grade: Pass with Distinction
Advisor: Prof. [Shay Cohen](#)
- 2011 – 2015 **B.E. in Electronics and Communication**, CMR Institute of Technology, Bangalore, India.
GPA: 7.58/10

Work Experience

- Oct 2021 – July 2022 **Applied Research Scientist - NLP**, Cookpad, Bristol, UK.
 - Operationalised an active learning, intermediate-task fine-tuning based multilingual meal-classification algorithm.
 - Implemented a label suggestion algorithm for efficient recipe tagging.
- May 2018 – Sept 2020 **Senior Software Developer - NLP**, Niki.ai, Bangalore, India.
 - Developed a real-time pause detection algorithm on speech signals to identify end-of-speech conditions.
 - Designed a probabilistic model to classify if an utterance is a valid parse or not.
 - Developed a bot that could respond to about 75 FAQs in regional languages and English.
 - Built a character-level RNN model to classify utterances as 'Hinglish' words with minimum lexicon support.
 - Created an algorithm to automatically generate contextual synonyms and augment concept-grammar data.
- Sep 2016 – Feb 2018 **Data Scientist - Research and Development**, Dave.ai, Bangalore, India.
 - Built a time series model to generate weekly/monthly sales forecasts of products with scarce data.
 - Devised a robust machine learning model to predict and optimize e-commerce metrics for retailers.
 - Created an image search engine to recommend similar images based on segmentation as a selective search.
- Feb 2016 – May 2016 **Junior Data Scientist**, Hotify.ai, Bangalore, India.
 - Aggregated tweets from pre-selected news sources to estimate trending hashtags.
 - Created actionable insights by analyzing how select twitter users through A/B testing react to real-time events.
 - Prepared training data by examining RSS feeds of news sources and rated each source using multiple factors.

Internships

- Aug 2024 – Feb 2025 **Research Intern**, Huawei R&D, Edinburgh, UK.
Topic: Integrating knowledge graphs in LLMs for language-to-code generation to solve math reasoning problems
Mentor: Prof. [Jeff Z. Pan](#)

- Aug 2021 – Oct 2021 **Research Assistant**, Usher Institute (University of Edinburgh) and Institute of Health Informatics (UCL), Edinburgh, UK.
Topic: Clinical Natural Language Processing for Automated Medical Coding [\[Details\]](#)
Mentor: Prof. [Honghan Wu](#) and Dr. [Hang Dong](#)
- May 2021 – Aug 2021 **Student Developer**, Google Summer of Code (GSoC), Remote.
Topic: Identify Joint Meaning Construal Patterns using Co-speech Gestures [\[Details\]](#) [\[Slides\]](#)
Mentor: Prof. [Tiago Torrent](#)

Selected Publications

- preprint Surfacing Program Structure with Auxiliary Signals for Low-Resource Code Translation
Federico Ranaldi, **Nickil Maveli**, Shay Cohen, Fabio Massimo Zanzotto
EMNLP 2026 (under review)
- preprint Are you Synthesizing or Recalling? Evaluating LLMs on Algorithmic Code Retrieval
Nickil Maveli, Antonio Vergari, Shay Cohen
EMNLP 2026 (under review)
- ACL 2026 (Findings) Can LLMs Compress (and Decompress)? Evaluating Code Understanding and Execution via Invertibility
[\[PDF\]](#)
Nickil Maveli, Antonio Vergari, Shay Cohen
Findings of ACL 2026
- NAACL 2025 (Findings) What can Large Language Models Capture about Code Functional Equivalence? [\[PDF\]](#)
Nickil Maveli, Antonio Vergari, Shay Cohen
Findings of NAACL 2025
- ACL 2022 (Findings) Co-training an Unsupervised Constituency Parser with Weak Supervision [\[PDF\]](#)
Nickil Maveli, Shay Cohen
Findings of ACL 2022
- WNUT@EMNLP 2020 EdinburghNLP at WNUT-2020 Task 2: Leveraging Transformers with Generalized Augmentation for Identifying Informativeness in COVID-19 Tweets [\[PDF\]](#)
Nickil Maveli
Proceedings of the 2020 EMNLP Workshop W-NUT
- Master's Thesis Co-training a Weakly Supervised Constituency Parser of Natural Language [\[PDF\]](#)
Nickil Maveli

Awards

- 2026 Unite! scholarship to attend LxMLS 2026
- 2025 Student Volunteer Award at NAACL 2025
- 2022 Student Volunteer Award at ACL 2022
- 2022 Fully-funded UKRI CDT in NLP scholarship to pursue PhD studies

Invited Talks

- June 2026 Invited Talk at Sapienza University of Rome and the University of Rome Tor Vergata
- July 2025, June 2026 Student Panel for MSc Dissertation Q&A event
- July 2022 International Society for Gesture Studies (ISGS), Symposium IV [\[Details\]](#) [\[Video\]](#)
- Dec 2021 University of Oxford, IMCC seminar [\[Details\]](#) [\[Video\]](#)
- Dec 2017 Techfest IIT Bombay, Artificial Intelligence summit [\[Details\]](#)

Technical Skills

- Specialties Natural Language Processing, Machine Learning, Deep Learning, Reinforcement Learning, Recommender Systems, Time-series Forecasting

Languages Python, R, Matlab, Javascript, SQL, HTML/CSS, L^AT_EX
Tools and Frameworks Pandas, NumPy, SciPy, Matplotlib, Scikit-Learn, NLTK, Keras, PyTorch, TensorFlow, PySpark, Flask, Django, Docker, Kubernetes, AWS, Jira, Git, Elastic Stack, MongoDB

Key Achievements

- 2026 Selected to participate in Lisbon Machine Learning Summer School (LxMLS 2026).
- 2025 Selected to participate in Advanced Language Processing School (ALPS 2026).
- 2022 Selected to audition as a DataCamp instructor for the “Introduction to mlflow” course.
- 2022 Secured Top 3% in the Evaluating Student Writing Challenge hosted on Kaggle; Won a silver medal. [\[Details\]](#)
- 2021 Selected to participate in the Lisbon Machine Learning School (LxMLS 2021). [\[Details\]](#)
- 2020 Secured Top 2% in the Jigsaw Multilingual Toxic Comment Classification Challenge hosted on Kaggle; Won a silver medal. [\[Details\]](#)
- 2019 Finished 2nd out of 479 competitors in the Quality Forecasting in Cement Manufacturing Challenge hosted on CrowdAnalytix; Won \$2,500 as prize money; Winner representing India. [\[Details\]](#)
- 2019 Secured Top 3% in the Quora Insincere Questions Classification Challenge hosted on Kaggle; Won a silver medal. [\[Details\]](#)
- 2018 Finished 2nd out of 750 competitors in the Predicting How Points End in Tennis Challenge hosted on CrowdAnalytix; Won \$2500 as prize money; [Featured in Tennis Australia's official press release.](#) [\[Details\]](#) [\[Media Coverage\]](#)
- 2017 Placed 3rd out of 2975 competitors in the Click Prediction Challenge hosted on AnalyticsVidhya; Awarded a cash prize of ₹25000. [\[Details\]](#)
- 2016 – present Among Top Stack Overflow Users for tags—Python, Pandas, NumPy, Matplotlib, with an overall 25000+ reputation score and more than 5 million post views worldwide. [\[Details\]](#)

Academic Services

Teaching Assistant

- 2025-26 Advanced Topics in Natural Language Processing (INFR11287) - Demonstrator, Marker
- 2024-25 Natural Language Understanding, Generation, and Machine Translation (INFR11157) - Tutor
- 2023-24 Cohort '22 Representative, CDTalks Organizer
- 2023-24 Accelerated Natural Language Processing (INFR11125) - Demonstrator
- 2022-23 Accelerated Natural Language Processing (INFR11125) - Tutor, Demonstrator, TA (Piazza)

Reviewer

- 2025-current COLM
- 2022-current ACL, NAACL, EMNLP
 - 2021 ACL, AmericasNLP
 - 2020 W-NUT@EMNLP (Shared Task 2), Scipy (Tutorials, Talk & Poster Presentations)

Volunteer

- 2021 ACL, ICML, NAACL, ICLR, EACL
- 2020 EMNLP

Respondent

- 2020 Semantic Scholar Diary Study project conducted by the Allen Institute for AI

Open Source Contributions

- 2022 – present ivy [\[Commits\]](#)